



ROCHESTER INSTITUTE OF TECHNOLOGY

COLLEGE OF SCIENCE

SCHOOL OF MATHEMATICS AND STATISTICS

STAT 547

PRINCIPLES OF STATISTICAL DATA MINING

Homework 1

THE PHILOSOPHY OF LEARNING BEFORE DOING

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Spring 2026

Assigned: February 5, 2026

Due: February 16, 2026, 11:59 PM EST

“The formulation of a problem is often more essential than its solution.”
- Albert Einstein

Assignment Philosophy

This assignment emphasizes **conceptual depth over computational breadth**. You will spend 80% of your time thinking and 20% computing. Each question is designed to reveal the philosophical underpinnings of statistical learning in the age of AI. You are expected to turn in the following: (a) **PDF Document** Professionally typeset using LaTeX/RMarkdown (b) **Concise Answers** Long and unclear answers will lose points, but extremely short and cryptic answers will not be accepted either. You must show your work and reasoning with just enough words. (d) **Mathematical Rigor** Proper notation and derivations are expected; (e) **Conceptual Depth** Genuine understanding, not memorization is the gold standard here.

PART 1: THE NINE WHEELS FRAMEWORK (20 POINTS)

Question 1: The Wheel of Problem Formulation (10 points)

A healthcare company wants to predict patient readmission within 30 days.

- **a)** Define this problem using the Nine Wheels terminology: $\mathcal{X}$ (input space), $\mathcal{Y}$ (output space), $\mathcal{H}$ (hypothesis space), ℓ (loss function).
- **b)** Explain why **precision** might be more important than accuracy here, using the business impact framework from Wheel 9.
- **c)** Describe what **data veracity** ($\mathbb{1}_{\{\text{data is reliable}\}}$) means in healthcare and three specific challenges.

Question 2: The Wheel of Theoretical Guarantees (10 points)

Consider the k-Nearest Neighbors (k-NN) algorithm for classification.

- **a)** Define the **Bayes classifier** $f^*(\mathbf{x})$ and its associated **Bayes risk**. Why is it the theoretical gold standard?
- **b)** State the universal consistency result for k-NN. What conditions on k and n are required for the k-NN risk to converge to the Bayes risk?
- **c)** Connect this result to the **bias-variance tradeoff**. How does the choice of k control this tradeoff in practice, even for finite n ?

PART 2: K-NEAREST NEIGHBORS LEARNING (20 POINTS)

Question 3: The Distance Perspective (10 points)

a) The performance of k-NN is highly sensitive to the choice of distance metric. For a feature vector $\mathbf{x} = (\text{age, weight, cholesterol})$, propose and justify a specific distance metric. Assume age is in years, weight in kg, and cholesterol in mg/dL.

- b) Explain the concept of the **curse of dimensionality** in the context of k-NN. How does the distribution of nearest-neighbor distances change as the number of features p grows?
- c) Describe two preprocessing techniques (from Wheel 3) that are **essential** for effective k-NN learning and explain why.

Question 4: The Model Complexity Perspective (10 points)

- a) In k-NN classification, how does the value of k directly control the **complexity** of the hypothesis space $\mathcal{H}$? Is a 1-NN model a high-capacity or low-capacity model? Justify your answer.
- b) Sketch two plots: (i) model complexity (y-axis) vs. k (x-axis), and (ii) expected prediction error (y-axis) vs. k (x-axis). Label the regions of underfitting and overfitting.
- c) Contrast k-NN (a *lazy*, instance-based learner) with a parametric model like logistic regression (an *eager*, model-based learner). List one significant advantage and one significant disadvantage of each paradigm.

PART 3: CONTRASTING LEARNING PARADIGMS (30 POINTS)

Question 5: Frequentist Gaussian Discriminant Analysis (15 points)

Consider Gaussian Discriminant Analysis (GDA) for a two-class classification problem, C_1 and C_2 , with class-conditional densities $p(\mathbf{x}|C_k) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma})$.

- a) Derive the log-odds $\log\left(\frac{P(C_1|\mathbf{x})}{P(C_2|\mathbf{x})}\right)$ in terms of the model parameters $(\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}, \pi_1, \pi_2)$. Show that the decision boundary is a linear function of $\mathbf{x}$.
- b) Write down the closed-form expressions for the Maximum Likelihood Estimates (MLE) of the parameters $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$, and the shared covariance matrix $\boldsymbol{\Sigma}$.
- c) Interpret the MLE for $\boldsymbol{\Sigma}$. How does it pool information from both classes, and what is a key **statistical** advantage of this pooling compared to having separate covariances?

Question 6: Computational Complexity and Scalability (15 points)

- a) Analyze the computational complexity of the standard k-NN algorithm for making a single prediction. Break it down into the complexity of (i) calculating distances and (ii) finding the k smallest distances.
- b) The standard k-NN algorithm has a time complexity of $O(nd)$ for a single prediction, where n is the training set size and d is the number of dimensions. Explain why this does **not** scale well for real-time applications with large n and d .
- c) Research and describe **one** algorithmic strategy (e.g., KD-Trees, Ball Trees, approximate nearest neighbors) used to accelerate the nearest neighbor search. Explain its core idea and under what conditions it is most effective.

PART 4: COMPUTATIONAL DEMONSTRATION (20 POINTS)

Computational exploration of kNN (10 points)

Implement the following simulation to illuminate the bias-variance tradeoff in k-NN, and then **answer the following question**: What does this visualization teach about the bias-variance tradeoff in k-NN? How does this connect to Question 4?

```
demonstrate_knn_tradeoff <- function() {
  set.seed(747)
  n <- 100
  # Generate simple non-linear data (e.g., sine wave + noise)
  x <- runif(n, 0, 2*pi)
  y <- sin(x) + rnorm(n, sd=0.3)
  test_x <- seq(0, 2*pi, length.out=100)
  true_y <- sin(test_x)

  # TODO: Fit k-NN models for k = 1, 5, 10, 20, 50, 100
  # Hint: use the 'knn' package or 'FNN::knn.reg()'

  # TODO: Plot the data and overlay the predictions for each k
  # TODO: Also plot the true function (sin(x))

  # TODO: Write a comment: For which k does the model overfit? Underfit?
  # Which k provides the best balance? How does this visualize the tradeoff?
}
```

Computational comparison between kNN and GDA (10 points)

Implement the following simulation to contrast k-NN and GDA on a simple dataset, and then **answer the following question**: Compare and contrast the decision boundaries. What does this tell you about the inherent assumptions of k-NN vs. GDA?

```
contrast_models <- function() {
  set.seed(747)
  # Generate data from two Gaussian classes with SOME overlap
  n <- 50
  class_1 <- MASS::mvrnorm(n, mu = c(0,0), Sigma = matrix(c(1, 0.5, 0.5, 1), 2))
  class_2 <- MASS::mvrnorm(n, mu = c(1.5,1.5), Sigma = matrix(c(1, 0.5, 0.5, 1), 2))
  data <- data.frame(rbind(class_1, class_2), class = factor(rep(c(1,2), each=n)))
  names(data) <- c("X1", "X2", "Class")

  # Create a grid for plotting decision boundaries
  grid <- expand.grid(X1 = seq(-3, 4.5, length=100), X2 = seq(-3, 4.5, length=100))

  # TODO: Fit a k-NN model (e.g., k=5) and predict on the grid
  # TODO: Fit a Linear Discriminant Analysis (LDA) model (special case of GDA) and predict on the grid
  # TODO: Plot the data points and the decision boundary for each model
  # Use contour plots or colored grids to show the predicted class regions
  # TODO: Write a comment: How do the decision boundaries differ?
  # Which model's assumption is more correct for this data? How can you tell?
}
```

Part 5: Video Presentation (10 points)

Create a 5-minute video presentation addressing: **Content Requirements:**

- Explain one key insight from Questions 1-6 that surprised you (e.g., the universal consistency of k-NN, the curse of dimensionality, or the ethical implications).
- Connect the theoretical guarantee of k-NN (from Q2) to the practical challenge of the curse of dimensionality (from Q3).
- Discuss what “AI-aware” statistics means to you personally.
- Propose one way this course could better prepare students for the AI future.

Technical Requirements: 5 minutes maximum (± 30 seconds); Clear audio and video quality; Professional presentation style; Your face and your slides must be both visible throughout; Upload as **private YouTube video** with access granted to: `epfega@rit.edu`; Include link in your submission

Grading Rubric: Conceptual clarity (4 points); Connection to course themes (2 points); Originality of insight (2 points); Presentation quality (2 points)

Part 6: Idiosyncratic (Mandatory, Ungraded)

This section is mandatory but will not contribute to your numerical grade. Its purpose is to encourage self-assessment and provide me with feedback to improve the course. Your thoughtful engagement is highly valued.

- (a) In your own words, explain the concepts from the first few weeks of STAT 747 that you feel you have truly mastered. What is the most challenging concept or practical skill (e.g., R coding, theory, interpretation) that you are still working to solidify?
- (b) After completing this homework assignment, what grade (e.g., A, B, C, etc.) do you *estimate* you will earn? Justify your estimate by referencing which exercises you found straightforward and which you found most challenging.

Submission Instructions

- Submit PDF and video link via **myCourses.rit.edu**
- PDF must include: Written answers + code + video link
- Video must be accessible until grading is complete
- Late policy: 10% deduction per day, maximum 3 days late